

The Dissipating Universe, Gradient Descent Rules

The MITC Model — A New Physical Framework Where Energy Gradients Drive Everything, Without Exception

Model Name: MITC Model (Multiplicative-Entropy-Driven Invariant-Topology Chiral-Space Model)

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What if space itself is the only reality? For a century, physics has been built on two incompatible pillars: General Relativity, which describes space as smooth, curving geometry, and Quantum Field Theory, which describes particles as excitations of abstract fields. Despite their mathematical elegance, neither tells us what space actually is, nor why time flows in only one direction. The MITC Model (Multiplicative-Entropy-Driven Invariant-Topology Chiral-Space Model) offers a different answer: space is a discrete network of fundamental units, connected like a fishing net that never breaks or rewires. Every physical phenomenon — motion, light, gravity, mass, even life — is simply energy redistributing itself across this fixed network. Energy flows only downhill, from higher-energy nodes to lower-energy neighbors, in discrete quanta. That single rule generates everything we observe.

The model's foundational dynamical postulate is remarkably simple: energy transfers irreversibly from higher-energy Space Elementary Quanta nodes to adjacent lower-energy nodes, preferentially along the steepest energy gradient. This rule admits no counterexamples — not in classical mechanics, not in thermodynamics, not in quantum systems. Every known process can be understood as gradient-driven dissipation: (1) potential energy gradients drive motion in Newtonian mechanics; (2) temperature gradients drive heat flow in thermodynamics; (3) metric curvature gradients drive geodesic motion in General Relativity; (4) electric potential gradients drive charge motion in electrodynamics; (5) even in quantum processes, apparent "unitary evolution" relies on external energy inputs such as lasers and magnetic traps to maintain coherence — when these external devices are included as part of a closed system, dissipation remains universal. No observed phenomenon supports reversible evolution in any closed physical system. This universality suggests that irreversibility is not an approximation, but a primitive feature of physical reality.

The MITC Model offers a unified explanation for a wide range of fundamental questions. On the nature of time: time is not a mysterious river, but the counting of discrete energy transfers across the network. Global cosmic time ticks with every network state update; local time dilation occurs because spatial deformation lowers the resonance frequency of elementary quanta — fewer transfers per external tick means time genuinely slows down in that region. On entropy: traditional entropy is a

statistical blur, whereas this model introduces multiplicative entropy — the product of all node energies. With every single energy transfer, this product increases; it never decreases. When energy becomes perfectly uniform, entropy reaches its maximum. This is the Second Law of Thermodynamics written not as a tendency, but as an exact, step-by-step mathematical fact. On mass and gravity: mass is not mysterious "stuff," but elastic energy stored in locally compressed regions of the spatial network, locked in place by a Higgs-like chiral mechanism. $SU(3)$ color forces compress the network while preserving spherical symmetry; the surrounding stretched space manifests as gravity. Mass and gravity are dual expressions of spatial elasticity. On dark matter and dark energy: dark matter is weakly compressed locked states — compressed but insufficient to form ordinary mass; dark energy is the ground-state energy of the spatial network itself. Both emerge naturally from the model's core assumptions. On the constancy of light speed: light is the maximum propagation speed of excitation waves across the spatial network; nothing can outrun the network's own limit. This is why velocities never add beyond c — not a postulate, but a consequence of the network's structure. On wave-particle duality and uncertainty: wave nature is fundamental; particle nature emerges from spacetime discreteness. Uncertainty arises because localizing to a single quantum loses information about propagation dynamics.

The MITC Model also makes a falsifiable prediction: a small but measurable difference exists between the magnetic moments of the positron and the electron, arising from their opposite chiral coupling to the fixed-spin SEQ ground state.

The MITC Model provides a unified and physically intuitive picture: no non-local interactions are required; global energy conservation is automatic, as adjacency relations never change; the arrow of time is built into the structure of space; quantum probability emerges from the coexistence of multiple equivalent maximum-entropy paths; life itself is understood as a system that records and invokes cumulative entropy-maximizing pathways.

The universe is not a clockwork mechanism winding down. It is a dissipative network, step by step, gradient by gradient, irreversibly homogenizing energy toward thermal equilibrium — without exception. This model does not negate existing physics; it simply asks: what if the mathematics points to a physical reality we have yet to visualize?